

Quantitative mapping of glycosaminoglycan secondary structure driven by sulfation and iduronic acid

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Glycosaminoglycans such as heparin and heparan sulfate are long, highly charged polysaccharides whose shape governs their binding to proteins. We map their secondary structure with all-atom molecular dynamics and enhanced sampling. The ring puckering of iduronic acid switches between the chair and skew-boat conformations, and sulfation at the 2-O position stabilizes the skew-boat form. Calcium ions bridge neighbouring sulfate groups and stiffen the chain.

Simulation details

Chains of twelve disaccharide units were simulated with the GLYCAM06 force field and the electronic continuum correction for ions, using OPC water at 300 K. Ring puckering was described by Cremer-Pople coordinates and sampled with metadynamics. Persistence length was computed from the decay of bond vector correlations.